Available online at www.sciencedirect.com

ScienceDirect

Physics of Life Reviews 15 (2015) 130–132

PHYSICS of LIFE reviews

www.elsevier.com/locate/plrev

Comment

Causal influence in neural systems: Reconciling mechanistic-reductionist and statistical perspectives Comment on "Foundational perspectives on causality in large-scale brain networks" by M. Mannino & S.L. Bressler

John D. Griffiths

Received 4 November 2015; accepted 4 November 2015

Available online 10 November 2015

Communicated by L. Perlovsky

Keywords: Causality; Neural mass modelling; EEG; MEG; fMRI; Neuroimaging

The modern understanding of the brain as a large, complex network of interacting elements is a natural consequence of the Neuron Doctrine [1,2] that has been bolstered in recent years by the tools and concepts of connectomics. In this abstracted, network-centric view, the essence of neural and cognitive function derives from the flows between network elements of activity and information – or, more generally, causal influence. The appropriate characterization of causality in neural systems, therefore, is a question at the very heart of systems neuroscience.

Accordingly, the past two decades has seen a substantial amount of neuroscientific research concerned with the development and application of analysis methodologies for estimating causal relations in neural systems from various kinds of neurobiological recordings. These may be grouped into two broad classes: *phenomenological* approaches and *physiological* approaches.[1] Phenomenological approaches, such as time- and frequency-domain Granger causality (GC; [3–5]); directed transfer function [6], partial directed coherence [7], structural equation modelling [8,9], and Bayesian networks [10,11], seek to describe relationships between observed neurobiological variables directly, generally using off-the-shelf algorithms developed in the fields of machine learning and applied statistics. Physiological approaches, in contrast, begin with a (typically relatively coarse-grained) biophysical generative model of the system in question, and seek to estimate parameters of that model using the measured data [12–20]. Causal influence is not so much the *raison d'être* of these models, but rather, as noted by Friston et al. [21], it is inherent in their mathematical formulation: fluctuations in (hidden) state variables such as the mean firing rate of a neuronal subpopulation induce changes in the state variables of other brain regions such as average post-synaptic membrane potentials, as specified ultimately by the model's equations of motion. The strengths of these causal influences between neuronal subpopula-

DOI of original article: http://dx.doi.org/10.1016/j.plrev.2015.09.002.

E-mail address: j.davidgriffiths@gmail.com.

[1] This distinction is one that is often made in comparisons of so-called 'functional' and 'effective' connectivity [15,25], where the latter is often equated with one specific family of techniques, dynamic causal modelling (DCM; [14]). The terminology used here is preferred as it is both more descriptive and more generic.

http://dx.doi.org/10.1016/j.plrev.2015.11.003

1571-0645/© 2015 Elsevier B.V. All rights reserved.

tions are determined by synaptic gain or coupling strength parameters, and are typically amongst the key parameters of interest that are optimized when fitting the model to experimental data.

The proposal by Mannino & Bressler [22] to regard causal relationships in large scale brain networks within the framework of probabilistic causality (PC) resonates well with the statistical philosophies motivating the various phenomenological approaches to causal connectivity outlined above. Causal weights – in terms of e.g. (conditional) probabilities, regression coefficients, and derivative quantities such as GC, all lend themselves naturally (and indeed, often preferentially) to probabilistic interpretations. It is less clear, however, how and whether physiologically-based approaches to modelling causal influences in brain networks might fit into the PC schema. In these models, connectivity strengths are generally taken to represent underlying physiological quantities such as synaptic densities, which connect the source and target populations through a mechanistic and (in principle) reductionist, rather than a statistical, relationship.[2] Indeed, it has been argued [21] that the key advantage and difference of physiological approaches over phenomenological approaches is that the former may be used to test mechanistic hypotheses, possibly arrived at through initial exploratory phenomenological analyses.

It should be noted that, whilst the emphasis thus far has been primarily from an empirical, data-analytic perspective, the questions raised go much deeper. This is because the models that inform the physiological approaches discussed above are not only statistical data analysis tools, but also constitute some of the key building blocks in several mathematical theories of brain function [22–24]. A truly foundational concept of causal influence in neural systems should apply equally to both theoretical structures and to data analysis procedures.

There are a number of ways in which Mannino & Bressler might address this challenge to their PC thesis. One may be to simply interpret connection strengths in physiological models as probabilistic by fiat, perhaps by appealing to the argument that the underlying phenomena are themselves inherently probabilistic, even if our necessarily over-simplified mathematical representations of them are not. This of course sidesteps the issue entirely, and would only really be adequate if the problem is not considered to be a serious one at either an applied or foundational level. More compelling would be to develop a formulation of PC that incorporated more directly the mechanistic causal relationships inherent in the physiological approach.[3] A third alternative could be to abstain entirely from PC interpretations of physiological model parameters, and highlight instead the difference between mechanistic-physical causal relationships and probabilistic causal relationships. The resultant plurality of causal influence ontologies may be less elegant than a single all-encompassing framework, but is not without precedent (see e.g. Cartwright [25]). Seth and colleagues [5] also argue for a distinction between the causal interpretations offered by one family of physiological models (DCM) and GC, suggesting that the two approaches "ask and answer fundamentally different questions, so that choosing one or the other (or both) depends on whether one is interested in describing the data in terms of information flow (GC) or exposing the underlying physical-causal mechanism (DCM)."

To conclude, an admirable feature of the target article is its synthesis of traditional and modern philosophical treatments of causality, together with current statistical and neurobiological theory and practice. Like most interesting ideas, the authors' characterization of interactions in large-scale brain networks within the framework of PC raises many questions, but with further refinement promises to provide a compelling foundational perspective on the functional organization of neural systems.

Acknowledgements

JG is supported by a J.S. McDonell Foundation Collaborative Research Grant, #220020255.

References

[1] Bullock TH, Bennett MVL, Johnston D, Josephson R, Marder E, Fields RD. Neuroscience. The neuron doctrine, redux. Science 2005;310(5749):791–3.

[2] Shepherd GM. Foundations of the neuron doctrine. New York: Oxford University Press; 1991.

[2] This is essentially a neuro version of the classical Russellian objection to causal interpretations in physics.

[3] It may be noted in connection with this that one of Salmon's chief concerns regarding the PC models of Reichenbach, Good, and Suppes, is their "attempt to carry out the construction of causal relations on the basis of probabilistic relations among discrete events, without taking account of the physical connections among them"' [26, pg. 66].

[3] Ding M, Bressler SL, Yang W, Liang H. Short-window spectral analysis of cortical event-related potentials by adaptive multivariate autoregressive modeling: data preprocessing, model validation, and variability assessment. Biol Cybern 2000;83(1):35–45.
[4] Brovelli A, Ding M, Ledberg A, Chen Y, Nakamura R, Bressler SL. Beta oscillations in a large-scale sensorimotor cortical network: directional influences revealed by Granger causality. Proc Natl Acad Sci USA 2004;101(26):9849–54.
[5] Seth AK, Barrett AB, Barnett L. Granger causality analysis in neuroscience and neuroimaging. J Neurosci 2015;35(8):3293–7.
[6] Kamiński MJ, Blinowska KJ. A new method of the description of the information flow in the brain structures. Biol Cybern 1991;65(3):203–10.
[7] Baccalá LA, Sameshima K. Partial directed coherence: a new concept in neural structure determination. Biol Cybern 2001;84(6):463–74.
[8] McIntosh AR, Gonzalez-Lima F. Structural equation modeling and its application to network analysis in functional brain imaging. Hum Brain Mapp 1994;2(1–2):2–22.
[9] Penny WD, Stephan KE, Mechelli A, Friston KJ. Modelling functional integration: a comparison of structural equation and dynamic causal models. Neuroimage 2004;23:S264–74.
[10] Smith SM, Miller KL, Salimi-Khorshidi G, Webster M, Beckmann CF, Nichols TE, et al. Network modelling methods for FMRI. Neuroimage 2011;54(2):875–91.
[11] Ramsey JD, Hanson SJ, Hanson C, Halchenko YO, Poldrack RA, Glymour C. Six problems for causal inference from fMRI. Neuroimage 2010;49(2):1545–58.
[12] Robinson PA, Rennie CJ, Rowe DL, O'Connor SC. Estimation of multiscale neurophysiologic parameters by electroencephalographic means. Hum Brain Mapp 2004;23(1):53–72.
[13] Kerr CC, Rennie CJ, Robinson PA. Physiology-based modeling of cortical auditory evoked potentials. Biol Cybern 2008;98(2):171–84.
[14] Friston KJ, Harrison L, Penny W. Dynamic causal modelling. Neuroimage 2003;19(4):1273–302.
[15] David O, Kiebel SJ, Harrison LM, Mattout J, Kilner JM, Friston KJ. Dynamic causal modeling of evoked responses in EEG and MEG. Neuroimage 2006;30(4):1255–72.
[16] Pinotsis DA, Moran RJ, Friston KJ. Dynamic causal modeling with neural fields. Neuroimage 2012;59(2):1261–74.
[17] Moran RJ, Kiebel SJ, Stephan KE, Reilly RB, Daunizeau J, Friston KJ. A neural mass model of spectral responses in electrophysiology. Neuroimage 2007;37(3):706–20.
[18] Valdes PA, Jimenez JC, Riera J, Biscay R, Ozaki T. Nonlinear EEG analysis based on a neural mass model. Biol Cybern 1999;81(5–6):415–24.
[19] Valdes-Sosa PA, Sanchez-Bornot JM, Sotero RC, Iturria-Medina Y, Aleman-Gomez Y, Bosch-Bayard J, et al. Model driven EEG/fMRI fusion of brain oscillations. Hum Brain Mapp 2009;30(9):2701–21.
[20] Aram P, Freestone DR, Cook MJ, Kadirkamanathan V, Grayden DB. Model-based estimation of intra-cortical connectivity using electrophysiological data. Neuroimage 2015;118:563–75.
[21] Friston K, Moran R, Seth AK. Analysing connectivity with Granger causality and dynamic causal modelling. Curr Opin Neurobiol 2013;23(2):172–8.
[22] Mannino M, Bressler SL. Foundational perspectives on causality in large-scale brain networks. Phys Life Rev 2015;15:107–23. http://dx.doi.org/10.1016/j.plrev.2015.09.002 [in this issue].
[23] Deco G, Jirsa VK, Robinson PA, Breakspear M, Friston K. The dynamic brain: from spiking neurons to neural masses and cortical fields. PLoS Comput Biol 2008;4(8):e1000092.
[24] Nunez PL. Toward a quantitative description of large-scale neocortical dynamic function and EEG. Behav Brain Sci 2000;23:371–98.
[25] Cartwright N. Where is the theory in our 'theories' of causality? J Philos 2006;103(2):55–66.
[26] Friston K. Causal modelling and brain connectivity in functional magnetic resonance imaging. PLoS Biol 2009;7(2):e33.